1- What are the breadth and the depth of DSP?

التغييرات الثورية

Revolutionary changes have already been made in a broad range of fields (breadth or applications):

- Communications
- medical imaging
- radar & sonar
- high fidelity music reproduction
- oil prospecting التنقيب عن الغاز

- Each of these areas has developed a deep DSP technology With its own:
  - algorithms.
  - mathematics &
  - specialized techniques.

→ This combination of breath & depth makes it impossible for any one individual to master all of the DSP technology that has been developed. يتقن رئيسي

2- What are the DSP education tasks?

involves two tasks:

① Learning general Concepts that apply to the field as a whole.

② Learning specialized techniques for your particular area of interest.

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3- Define the following items :

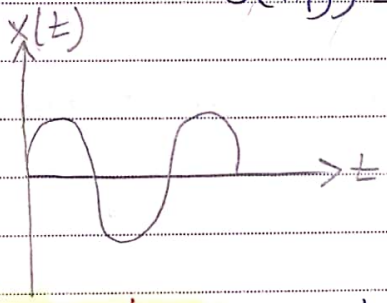
① Signal : is define as any physical quantity that varies with time, space or any other independent variable or variables.

$s_1(t) = 5t$, $s_2(t) = 20T^2 + 3t$.

$s(x,y) = 20x^2 + 2xy$.

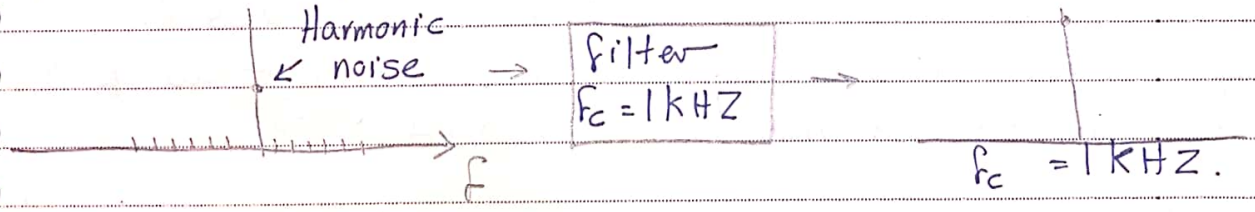
$x, y \rightarrow$ independent variables

$s(x,y) \rightarrow$ independ Variables.



$x(t) = A\sin(2\pi f t + \theta)$.

② System : is defined as aphysical device that performs an operation on a signal.



The filter that use to reduce the noise on the signal

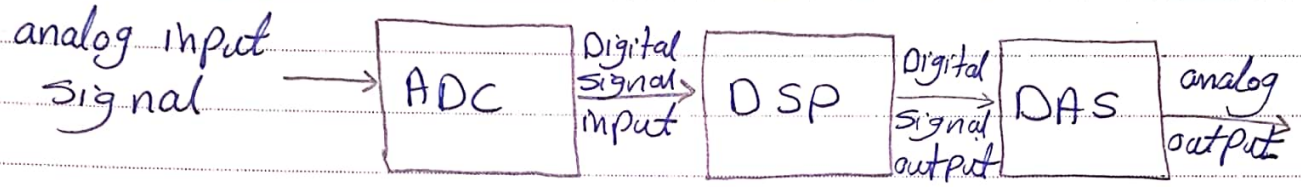
③ Signal processing : The set of operation that performs on the signal. there are two types of signal processing.

Ex : dsp , Asp

① analog signal processing (Asp)

analog Input Signal → [ASP] → analog Output Signal

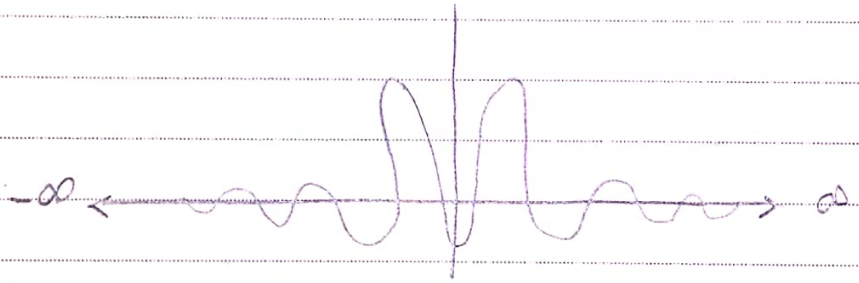
② Digital Signal processing (DSP)



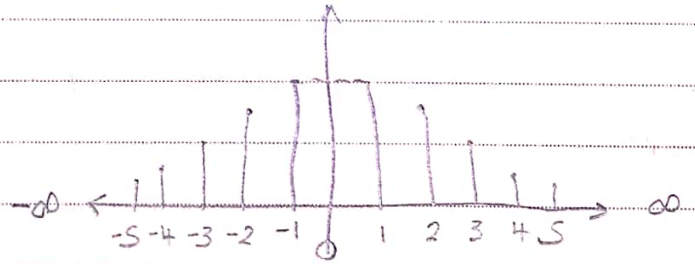
4- Draw & Explain the Classification of Signals.

① Continuous-time versus Discrete time signals:

Continuous time signals "analog" defined for every value of time (a, b) ⟶ a ----> $-\infty$, b ----> ∞



- Discrete time signals : Signal has value at samples of time.



② Continuous Valued Vs Disc. Valued Sig.s

→ The values of a Continuous-time or discrete time signal can be Continuous or discrete. If a signal takes on all possible values on a finite or an infinite range it is to be Continuous-Valued signal.

→ if the signal takes on values from a finite set of possible values, it is said to be a discrete-valued signal.

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③ Deterministic Versus Random Signals :

Any signal that can be uniquely described by an explicit mathematical expression, a table of data, or well-defined rule is called deterministic.

otherwise, we refer to these signals as random.

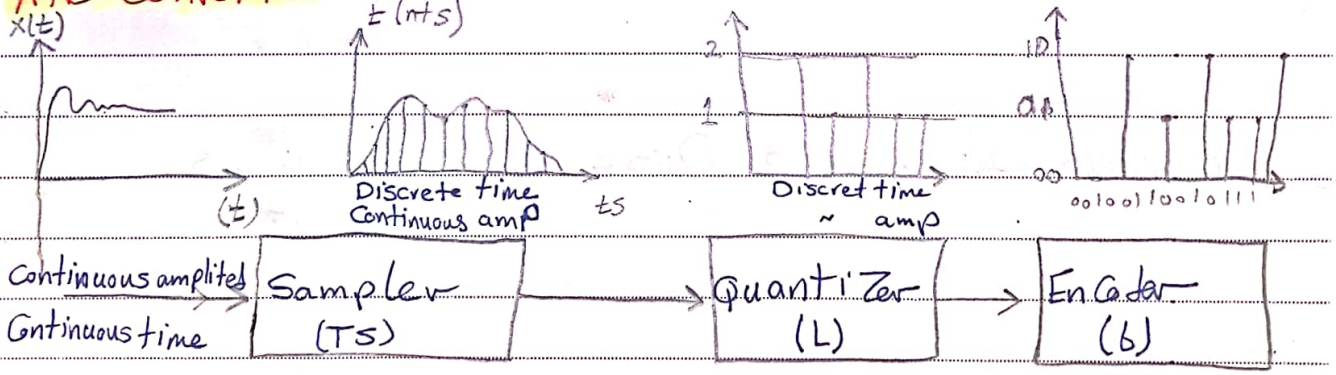
④ Multichannel & multidimensial signals.

$$s3(t) = \begin{bmatrix} s_1(t) \\ s_2(t) \\ s_3(t) \end{bmatrix}$$

We refer to such a vector of signals as a multichannel signal.

on the other hand, a signal is called M-dimensional if its value is a function of M independent variables.

5- Explain the basic three operations performed in A/D Converters.



Sampler → Converts Continuous time signal to Discrete time signal each Ts

Quantizer → Convert Continuous amplitude to discrete amplitude by reprenting each sample value to the nearest level

Difference between unquantized & quantized signal $eq(n) = x_q(n) - x(n)$

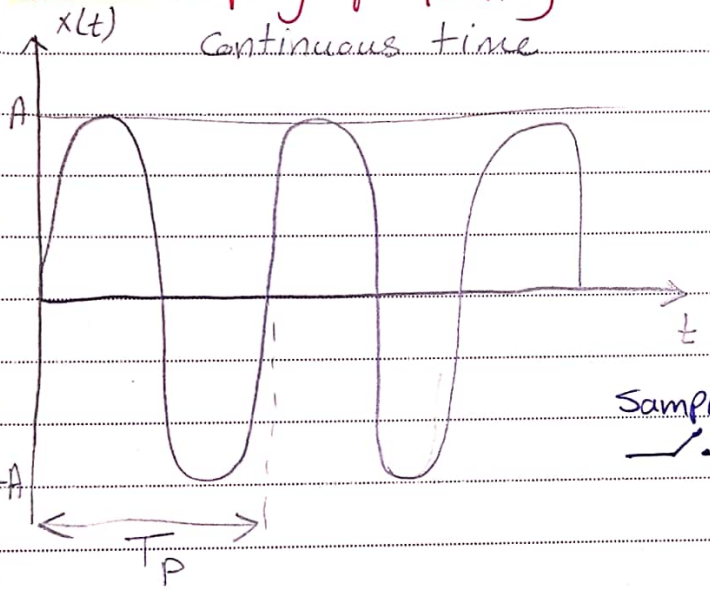
$$\Delta = \frac{Kmax - Kmin}{L-1}$$

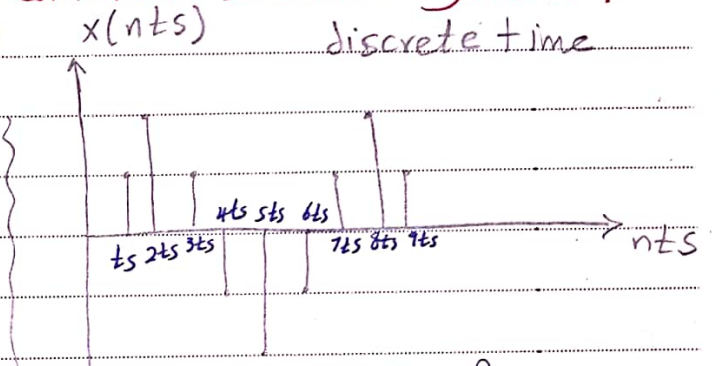
$eq(n)$

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Encoder → Represts each level by no of bits.

6- find the relation between the frequency of the discrete time signal, the frequency of Continuous time signals & the sampling frequency





Sampler

$x(nts) = A\sin(2\pi f nts + \theta)$

$x(nts) = A\sin(2\pi \frac{f}{fs} n + \theta)$

$x(t) = A\sin(2\pi f t + \theta)$

A : Amplitude.

θ : phase shift (degree)

f : freq $= \frac{1}{T_p}$ (cycle/sec = HZ)

t : foundmental period (sec) is periodic every T_p " T_p يتكرر "

$x(t)_f = \pm T_p$

$f_s = \frac{1}{T_s}$: Sampling freq (HZ = Sampling/sec).

$x(nts) = A\sin(2\pi nf + \theta)$.

$F = \frac{f}{f_s} \frac{\text{cycle} * \text{sec}}{\text{sec} * \text{sampling}} = \frac{\text{cycle}}{\text{sampling}}$

cycle يحتاج $F = \frac{1}{7}$ مثلا
7 Sampling في

f : digital frequency

: negative

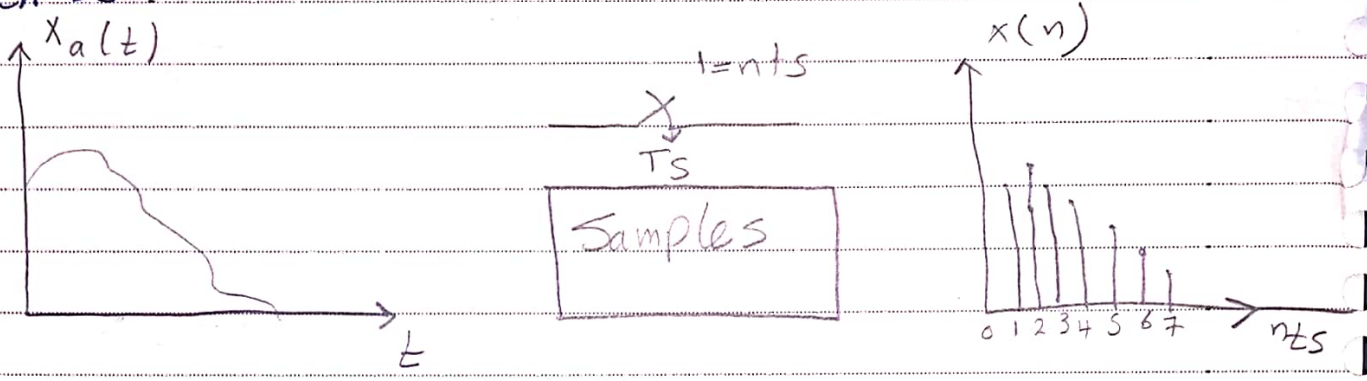
: Frequency of discrete time signal

~~Tp : Foundmental period~~

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7- State and explain Sampling theorem?

Convert Cntinous time signal to Discrete time signal each ts.

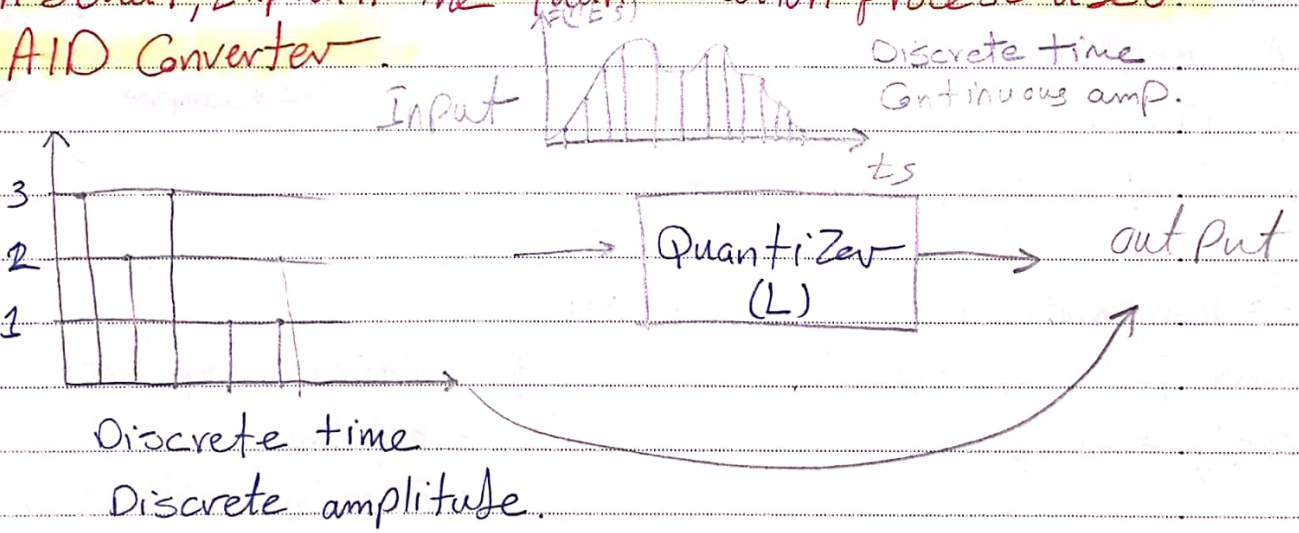


nyquist Condition $F_s \geq 2f$ (Sampling frequency; max freq.)

T_s = sampling time sec/sample.

F_s = freq $= \frac{1}{t_s}$ sample/sec.

8- In detail, Explain the quantization process used In A/D Converter.



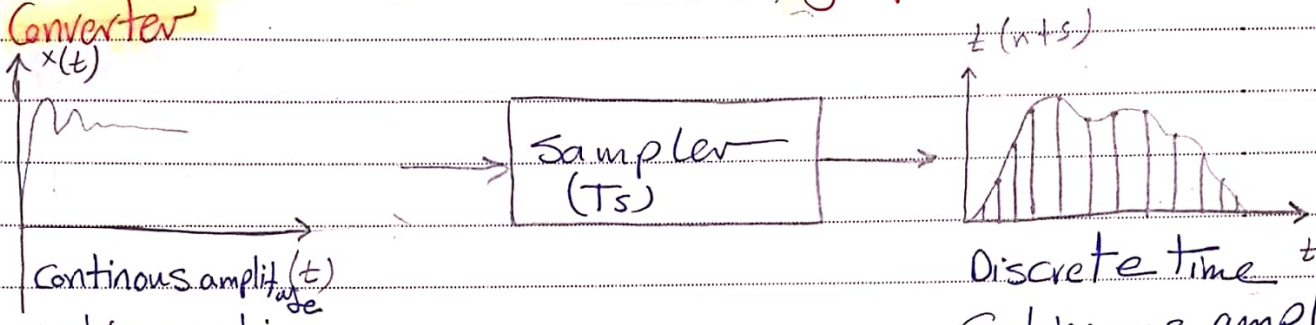
Quantizer → Convert Cntinuous amplitude to discrete amplitude by representing each sample value to the nearest level.

Difference between unquantized & quantized signal

$eq(n) = x_a(n) - x(n)$ $eq(n)$

$\Delta = \frac{x_{max} - x_{min}}{L-1}$

9- In Detail, Explain the Sampling process used in A/D Converter



Sampler → Converts Continuous time signal to Discrete time signal each T_s.

11- Define the items : i) Correlation operation: process which show 2 data sequence we can use Correlation to Compare to similarity

iii) Correlation Coefficient : used to measure cross correlation according to the fixed scale between -1 & +1.

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$$r_{12}(j) = \frac{1}{N}\sum_{n=0}^{N-1} x_1(n)\,x_2(n+j)$$

where x1 reference signal
x2 delayed output from crcuit

(8) j : response the amount of flag
N : no of samples

vi) Convolution operation : How the i/p interact with the sys to produce the o/p.

(ii) correlation theorem : Can be used to obtain the linear correlation by adding augmenting zeros to the 2 sequences.

13- For the LTIS define the following items:

i) Linearity:

A discrete time system is said to be linear if it obeys the principle of superposition. Ex: the response of a linear system to 2 or more i/p is equal to sum of the responses of the system to all the other i/p

if i/p = $x_1(n)$ ⟶ system o/p $y_1(n)$

$x_2(n)$ ⟶ system o/p $y_2(n)$

the response of the system to both i/p will be:

$a_1x_1(n) + a_2x_2(n) \longrightarrow a_1y_1(n) + a_2y_2(n)$

(where a_1, a_2 are arbitrary constants)

ii) Consality:

The system is used to be causal system if it produces an o/p only where there is an input. all physical systems are causal. In general a causal D.T system has zero impulse response before time zero ($h(k)=0$ for all $k<0$) also the D.T sequence say $x(n)$ is causal if its value is zero befor time zero ($x(n)=0$ for all $n<0$)

iii) stability:

An LTIs is stable if its impulse response $h(k)$, satisfied the following condition : $\sum_{k=-\infty}^{\infty} h(k) < \infty$

This condition is satisfied if, $h(k)$ is of finite duration. or if $h(k)$ delays towards zero as k increase.

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iv) RoC : Region of Convergence تقارب — تلاقي
it's the region where the Z-transform converges & in this region the values of $x(z)$ are finite

BiBo: "bounded i/p bounded o/p"
~~The regio~~ An LTI system is said to be BIBO stable if and only if it satisfies criterion

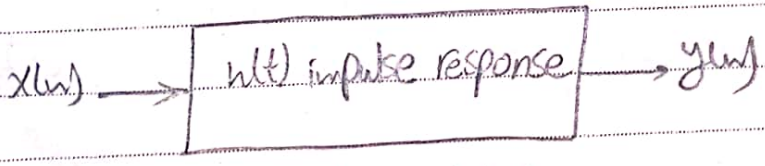
$$\sum_{k=0}^{\infty} |h(k)| < \infty$$

$h(k)$: impulse with response the system

Time invariant system: ثابت
A discrete time signal is said to be time invariat if it is output is independent of the time the i/p is applied

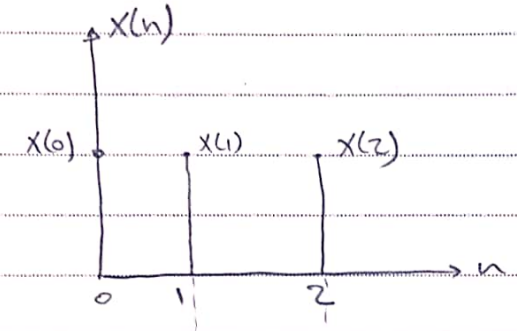
$$x(n) \longrightarrow y(n)$$
$$x(n-k) \longrightarrow y(n-k)$$

(12) Describe the convolution operation & deduce its compact relations.
Convolution operation : How the i/p interact with the system to produce the output



$$y(n) = x(n) * h(n)$$

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$y(0) = x(0)\,h(0)$

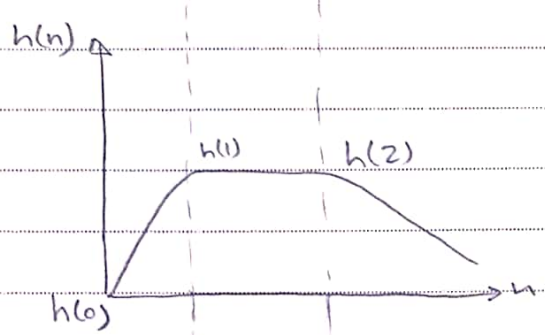
$y(1) = x(0)h(1) + x(1)h(0)$

$y(2) = x(0)h(2) + x(1)h(1) + x(2)h(0)$

$\vdots$

$y(n) = x(0)h(n) + x(1)h(n-1) + \dots + x(n)h(0)$

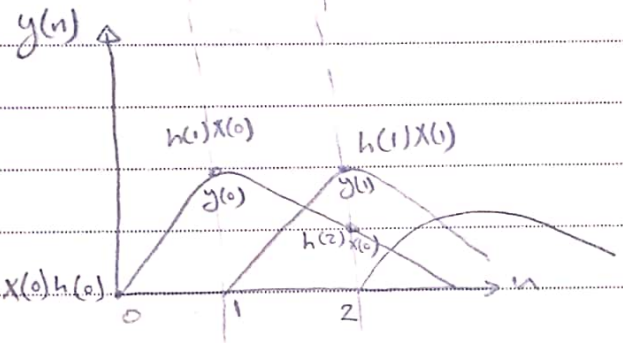
$= \sum_{m=0}^{n} x(m)\,h(n-m)$ #



OR

$y(n) = h(0)x(n) + h(1)x(n-1) + \dots + h(n)x(0)$

$= \sum_{m=0}^{n} h(m)\,x(n-m)$ #



Ques 2 True & false :

1) In quantization process, when the quantization step (Δ) increases, the accuracy of the quantizer increases — decrease ✗ — $\Delta = \frac{x_{max} - x_{min}}{L-1}$

2) Nyquist frequency (F_N) : is the minimum sampling frequency ✓

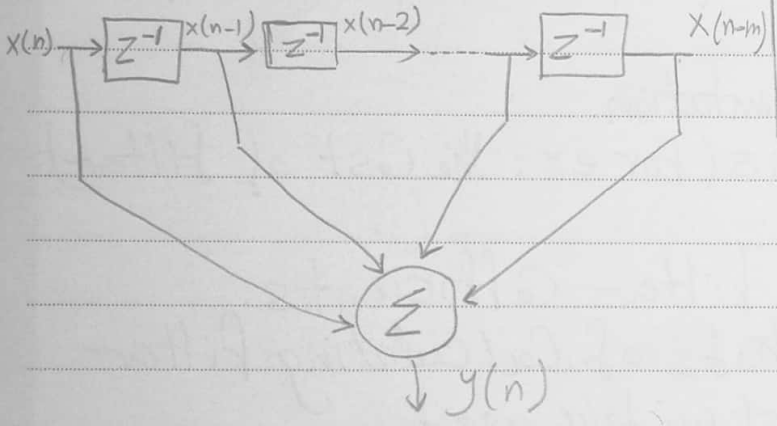
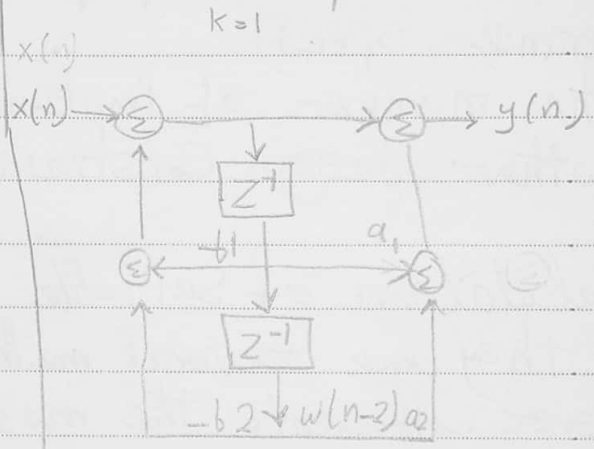
3) fourier transform (FT) is calculated for non periodic continuous-time signals. ✓

4) Discrete fourier transform (DFT) is obtained for analog signals. Digital ✗

5) to reconstract the continuous time signal $x(t)$ from its discrete time $x(n)$, F_s must be less than of $x(t)$ ✗

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→ Compare between FIR & IIR.

FIR	IIR
* finite impulse Response	* infinite impulse Response
$H(z) = \sum_{k=0}^{N-1} h(k) z^{-k}$	$H(z) = \frac{\sum_{k=0}^{N} a_k z^{-k}}{1 + \sum_{k=1} b_k z^{-k}}$
	
Difference eq $y(n) = \sum_{k=0}^{m} h(k) x(n-k)$	Difference eq: $w(n) = x(n) - b_1 w(n-1) - b_2 w(n-2)$
$y(n) = h(0)x(n) + h(1)x(n-1) \ldots + h(m)x(n-m)$	$y(n) = a_0 w(n) + a_1 w(n-1) + a_2 w(n-2)$
Delay ↓	Delay ↑
stability = always stable "No feedback"	stability = may stable & may not
Accuracy ↓	Accuracy ↑
transversal (direct)	direct form
frequency sampling	Cascade
fast Convolution	parallel
Lattice &	

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nce equation :

→ filter design steps :

1) specification of the filter requirements it include.
   i) signal charactristics (type of signal source and sink, I/O interface datarates & width & highest frequency of interest)
   ii) the charactristics of filter (Amplitude, phase response, tolerance, speed)
   iii) the manner of implementation.
   iv) other design Constrains (For ex: the Cost of filter).

2) Calculation of Suitable filter Cofficients.
   - they are several methods of Calculating filter ~~Coffiec~~ Cofficients the most widely used
   - Impulse invariant (IIR)
   - Window (FIR).
   - Bilinear transformation (IIR)
   - Frequency Sampling (FIR)
   - Pole-Zero pLacement (IIR)
   - Optimal (FIR)

3) representation of filter by suitable structure (realization)

FIR	IIR
direct	direct form
freq sampling	Cascade
fast Convolution	Parallel
Lattice	

4) Analysis of infite word length effects.
main Source of performance degradation in digital filters are
① input/output signal quantization
② Coffecient quantization ③ arithmatic round errors
④ over plow

5) implemetation of a filter.

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→ Difference equation :

specifies the actual operation that must be performed by the discrete time system on the i/p data in the time domain in order to generate the desired o/p.

it represents the relation between the system response with the i/p z- domain $y(n) = \sum_{K=0}^{N} a_K x(n-k) - \sum_{k-1}^{M} b_K$

$y(n-k)$

$x(n)$ ⟶ i/p sample

$y(n)$ ⟶ o/p sample

$y(n-k)$ ⟶ previous o/p

a_k, b_k ⟶ system Coefficients

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4) Prove that the DFT, $X(K)$ of sequence $X(n)$ is periodic

$$X(K) = \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi nK}{N}}$$

$$X(K+N) = \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi n(K+N)}{N}} = \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi nK}{N} + (-j2\pi n)}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi nK}{N}} \cdot e^{-j2\pi n} \quad \text{(1)} = \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi nK}{N}} = X(K)$$

$\therefore X(K) = X(K+N)$

$\therefore$ Sequence of $x(K)$ is periodic every N samples.